Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Ivan Jacob Agaloos Pesigan[1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]

[1]Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University

[2]Department of Biobehavioral Health, The Pennsylvania State University

[3]Department of Human Development and Family Studies, The Pennsylvania State University

Author Note

Ivan Jacob Agaloos Pesigan https://orcid.org/0000-0003-4818-8420; Michael A. Russell https://orcid.org/0000-0002-3956-604X; Sy-Miin Chow https://orcid.org/0000-0003-1938-027X.

This research was made possible by the Prevention and Methodology Training Program (PAMT) funded by a T32 training grant from the National Institute on Drug Abuse (NIDA).

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer.

Correspondence concerning this article should be addressed to Ivan Jacob Agaloos Pesigan, Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University, 320 Biobehavioral Health Building, University Park, PA 16802 or by email (ijapesigan@psu.edu).

Abstract

Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time-interval values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Mediation is a widely adopted model across the social, behavioral, and health sciences. A recent Google Scholar search, conducted as of 24 January 2024, using search terms `"indirect effect OR mediation AND social science OR behavioral science OR health science"`, yielded over five million entries. While many mediation models still rely on cross-sectional data, where time is ignored (Figure 1a), there is a growing trend toward employing longitudinal designs, where time is implicitly considered in the model (e.g., Figures 1b and 1c) addressing the critical issue of temporal ordering, thus, enhancing the plausibility of causal interpretations regarding mediation effects (Cole & Maxwell, 2003; Georgeson et al., 2023; Maxwell & Cole, 2007; Maxwell et al., 2011; O'Laughlin et al., 2018). Mediation models help us uncover the intermediate steps through which an independent variable X (such as an intervention) influences a dependent variable Y (such as an outcome) through one or more mediator variables M (e.g., Baron & Kenny, 1986; MacKinnon, 2008). By identifying these mediating variables, we gain insights into the direct and indirect causal pathways that connect interventions to their effects (Fairchild & MacKinnon, 2014; MacKinnon, 1994; Mackinnon & Dwyer, 1993; O'Rourke & MacKinnon, 2018, 2019).

The aim of this manuscript is to provide methods to quantify and visualize the uncertainty associated with point estimates of direct, indirect, and total effects as they change across a range of time-interval values using the continuous-time vector autoregressive (CT-VAR) framework. We do this with repeated measurements over time as a stronger basis for causal inference[1] in contrast to cross-sectional designs (Gollob & Reichardt, 1987, 1991). Furthermore, we take the continuous-time framework as previous

[1] For observational data, we limit our treatment of causality following the definition provided by Granger (1969), that is, when one variable, denoted as X, Granger-causes another variable, denoted as Y, it implies that past values of X contain information that helps predict future values of Y beyond what can be predicted using only past values of Y itself.

research (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) have shown its benefits particularly in addressing the time-interval dependency problem, that is, direct, indirect, and total effect sizes change as a function of the time-interval used to measure the variables. Our novel contributions to this area of research include introducing the delta and Monte Carlo methods as ways to quantify the uncertainty associated with point estimates of direct, indirect, and total effects across a range of time-intervals. Furthermore, we use the delta and Monte Carlo methods to generate regions of significance which help in visualizing for which range of time-interval values the direct, indirect, and total effects are statistically different from zero. Lastly, the methods we developed as well as visualization tools are made available in a free and open-source `R` package `cTMed`.

This manuscript is divided into the following sections: (1) introduction of continuous-time mediation and the need to quantify uncertainty associated with mediation effects; (2) hypothetical examples where we introduce point estimates for continuous-time mediation effects, delta and Monte Carlo method confidence intervals around point estimates, and regions of significance across a range of time-intervals; (3) simulation study examining the finite sample performance of the delta and Monte Carlo method confidence intervals; and (4) application of the delta and Monte Carlo method confidence intervals on an empirical data set. We close with a summary of our novel contributions, limitations of the current work, and future directions.

Mediation Using Longitudinal Data

In this section we discuss the rationale for moving from discrete-time to continuous-time mediation. We provide an overview of the continuous-time modeling framework, namely the linear stochastic differential equation model. While application of the continuous-time framework for mediation models exist in the literature (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021), quantifying the uncertainty associated with mediation effects are yet to be developed. The last part of this section outlines how we address this research gap.

From Discrete-Time to Continuous-Time Mediation

Mediation and causality have always been linked. From the "causal steps" approach (e.g., Baron & Kenny, 1986; James & Brett, 1984; Judd & Kenny, 1981) to the longitudinal framework (e.g., Cole & Maxwell, 2003; Gollob & Reichardt, 1987, 1991; Maxwell & Cole, 2007; Maxwell et al., 2011), the idea is that causality operates not only directly ($X \rightarrow Y$), but also indirectly ($X \rightarrow M \rightarrow Y$). Gollob and Reichardt (1987, 1991) proposed three principles of causality namely (1) "causes take time to exert their effects and, therefore, causal variables must occur before outcome variables"; (2) "variables can have effects in themselves"; (3) "size of the effect typically varies with the length of time lag".

Principles 1 and 2 underscore the fact that cross-sectional mediation presented in Figure 1a cannot capture the direct and indirect causal associations between X, M, and Y and the need to move to a model where previous values of variables are used to predict their current values as in the autoregressive cross-lagged mediation model presented in Figures 1b and 1c. The term autoregressive (AR) refers to allowing the variables to have effects in themselves and the term cross-lagged or crossregressive (CR) refers to using previous values of other variables to predict current values of a variable. The umbrella term used for these types of models is discrete-time vector autoregressive model (DT-VAR). The term vector autoregressive (VAR) means that there are more than one variable in the system. The term discrete-time (DT) means that we observe the variables at equally spaced time intervals (de Haan-Rietdijk et al., 2017). DT-VAR encompasses panel models where the observations are made in a few time points as well as data from intensive longitudinal data collection where many repeated observations are made.

Principle 3 underscores the time-interval dependency, that is, in DT-VAR, effect sizes change as a function of the time-interval used to measure the variables (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021; Wang & Zhang, 2020). This means that, given the same underlying causal process, the moment in time we used to observe the variables of interest (e.g., daily, monthly, yearly) influences the size of the effects we observe.

In a cross-sectional mediation, if X changes by one unit, then Y changes by τ' units, holding M constant. This is the direct effect of X on Y. If X changes by one unit, then M changes by α units as a result. Similarly, if M changes by one unit, then Y changes by β units as a result. The resulting change of α units on M produced by a unit change in X would then produce a corresponding change of β units on Y. Thus, Y changes by $\alpha\beta$ units through M as X changes by one unit. This is the indirect effect. The total effect is the sum of direct and indirect effects. In DT-VAR, this idea of decomposing effects also holds. However, we are tracing effects from previous values of the variables to the next. The indirect effect of X at time t on Y at time $t+2$ is the product of the path from X at time t to M at time $t+1$ and the path from M at time $t+1$ to Y at time $t+2$. While many mediation models still rely on cross-sectional data, the good news is that there is a growing trend toward employing DT-VAR models[2]. However, the indirect effects and all the other effects we observe in DT-VAR changes on account of the time-interval we use to observed the variables. A change in modeling framework from discrete-time to continuous-time can address this issue.

The challenges associated with DT-VAR can be addressed by adopting a continuous-time vector autoregressive (CT-VAR) modeling framework. CT-VAR parameters are defined at any time-interval, as such we can obtain the direct, indirect, and total effects for any time-interval. From this framework, we can explore how effects change as a function of time-interval. Additionally, unlike DT-VAR, which assumes equally spaced data points, CT-VAR accommodates unequally spaced data–a common occurrence in intensive longitudinal studies. These models allow us to capture the evolving dynamics of parameters across a range of time intervals (Boker, 2002; Oud & Delsing, 2010; van Montfort et al., 2018; Voelkle et al., 2012). However, embracing a continuous time perspective necessitates a conceptual shift. Specifically, lagged regression parameters, that

[2] See the decrease in cross-sectional studies from the systematic review by Maxwell and Cole (2007) to O'Laughlin et al. (2018).

is the AR and CR parameters, for a given time-interval within this framework are interpreted as total effects, rather than direct effects (Aalen et al., 2012, 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021).

The Linear Stochastic Differential Equation (SDE) Framework

In our modeling approach, we consider processes that continuously evolve over time. This means that the variables can assume values at any given moment. To capture this behavior, we employ linear stochastic differential equations (SDEs). Specifically, we focus on a special type of SDE known as the Ornstein–Uhlenbeck model (Chow et al., 2023; Oravecz et al., 2011; Uhlenbeck & Ornstein, 1930).

Similar to structural equation modeling, we can break down our model into two key components: the measurement component and the structural component. The measurement model describes the relationship between observed variables and latent (unobserved) variables. Specifically, consider the following equation:

$$\begin{aligned} \mathbf{y}_{i,t_m} &= \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \\ \text{with} \quad \boldsymbol{\varepsilon}_{i,t_m} &\sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Theta}\right) \end{aligned} \tag{1}$$

where $\mathbf{y}_{i,t_m}$ represents a vector of observed random variables, $\boldsymbol{\eta}_{i,t_m}$ a vector of latent random variables, and $\boldsymbol{\varepsilon}_{i,t_m}$ a vector of normally distributed random measurement errors, for time t at measurement occasion m and individual i. $\boldsymbol{\nu}$ denotes a vector of intercepts, $\boldsymbol{\Lambda}$ a matrix of factor loadings, and $\boldsymbol{\Theta}$ the covariance matrix of $\boldsymbol{\varepsilon}$. In the context of a mediation model with three variables where the effect the variable X on the variable Y is mediated by the variable M, we define $\mathbf{y}$ and $\boldsymbol{\eta}$ as follows:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix}. \tag{2}$$

The dynamic structure is given by

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = -\boldsymbol{\Phi}\left(\boldsymbol{\mu} - \boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t} \tag{3}$$

where $\boldsymbol{\mu}$ is the equilibrium or mean value, $\boldsymbol{\Phi}$ is the drift matrix, $\boldsymbol{\Sigma}$ is the matrix of volatility or randomness in the process (also known as the diffusion matrix), and $\mathrm{d}\boldsymbol{W}$ is a Wiener process or Brownian motion, which represents random fluctuations. The Ornstein–Uhlenbeck process centers around a central tendency denoted by $\boldsymbol{\mu}$. When the process deviates from this mean, it naturally gravitates back toward it. We can think of $\boldsymbol{\mu}$ as an "attractor" that pulls the process back to its typical value. When we specify the model such that the latent state variables have a stable mean vector $\boldsymbol{\mu}$ consisting of zeros, Equation 3 simplifies to

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = \left(\boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t}. \tag{4}$$

The diagonal elements of the matrix $\boldsymbol{\Phi}$ (referred to as auto effects) control the speed at which the process reverts to its own equilibrium value, $\boldsymbol{\mu}$. When the absolute value of these elements is high, the variable quickly returns to its mean after being perturbed. Smaller absolute values result in slower reversion, allowing the variable to exhibit more persistent behavior away from its mean. The off-diagonal elements of $\boldsymbol{\Phi}$ represent associations between variables and their rates of change. A negative cross effect indicates that if one variable takes on a positive value, there is a resulting decrease in another variable's value. The magnitude of this effect is determined by the absolute value. The diagonal elements of the matrix $\boldsymbol{\Sigma}$ determine the magnitude of volatility around the central tendency $\boldsymbol{\mu}$. Larger values indicate greater fluctuations or noise in the process, while smaller values result in smoother trajectories. Essentially, $\boldsymbol{\Sigma}$ quantifies the degree of randomness or shock-induced variability. The off-diagonal elements of $\boldsymbol{\Sigma}$ represent correlations between random shocks across variables in the system. In summary, the Ornstein–Uhlenbeck process is a mean-reverting stochastic process. Over time, it drifts

toward its equilibrium value $\boldsymbol{\mu}$, experiences volatility (as captured by $\boldsymbol{\Sigma}$), and reverts back to the mean with a rate determined by $\boldsymbol{\Phi}$. For the context of this manuscript, we refer to the Ornstein–Uhlenbeck model as the first-order continuous-time vector autoregressive model (CT-VAR).

The drift matrix $\boldsymbol{\Phi}$ holds particular significance in the context of mediation models. It serves as a conduit for understanding the associations between the states of the variables under investigation. Notably, within a trivariate model, involving η_X, η_M, and η_Y, numerous paths from one variable to another can be explored. This comprehensive connectivity arises because we can construct the model such that all variables are interconnected. Consequently, beyond the classic mediation model $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$, we can also investigate alternative mediation pathways, such as $\eta_Y \rightarrow \eta_M \rightarrow \eta_X$, using the same data set. For our current focus, we will delve into the specific $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ path. In the subsequent section, we will explore the relationship between CT-VAR and DT-VAR models, emphasizing the pivotal role played by the drift matrix $\boldsymbol{\Phi}$ within the $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ mediation model.

Link Between CT-VAR and DT-VAR

To understand the connection between CT-VAR and DT-VAR models, we need to examine the discrete-time state space model parameterization of the CT-VAR model which is the discrete-time solution in integral form given by

$$\begin{aligned} \boldsymbol{\eta}_{i,t_{m_i}} &= \boldsymbol{\alpha}_{\Delta t_{m_i}} + \boldsymbol{\beta}_{\Delta t_{m_i}} \boldsymbol{\eta}_{i,t_{m_i-1}} + \boldsymbol{\zeta}_{i,t_{m_i}}, \\ \text{with} \quad \boldsymbol{\zeta}_{i,t_{m_i}} &\sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}\right) \end{aligned} \tag{5}$$

where $\boldsymbol{\eta}_{i,t_{m_i}}$ and $\boldsymbol{\eta}_{i,t_{m_i-1}}$ represent the vector of latent variables in the current and previous measurement occasion m while $\boldsymbol{\zeta}_{i,t_{m_i}}$ is the normally distributed error term in the current measurement occasion. The model parameter $\boldsymbol{\alpha}$ denotes a vector of intercepts, $\boldsymbol{\beta}$ a matrix of coefficients describing the impact of the lagged values $\boldsymbol{\eta}_{i,t_{m-1}}$ on the current value $\boldsymbol{\eta}_{i,t_m}$ (AR coefficients on the diagonal and CR coefficients on the off-diagonal), and $\boldsymbol{\Psi}$ the

covariance matrix of $\boldsymbol{\zeta}$. Notice that the parameters have subscript Δt, representing that their values are dependent on the time-interval, that is, given the same continuous process, their values change depending on the time-interval.

In the context of CT-VAR models, the underlying process can take values at any arbitrary time point. However, our observations occur discretely. This means that while the process itself operates continuously, we only observe it at specific discrete time instances. Let's compare this to DT-VAR, where observations are made at specific discrete time intervals (usually positive integers). In a DT-VAR example, suppose we collect data every hour from 7 AM to 10 AM for four measurement occasions. We make discrete observations in time t, scaled by the hour, on specific measurement occasions $m = \{1, \cdots, 4\}$. These occasions correspond to time points: $t_1 = 7$, $t_2 = 8$, $t_3 = 9$, and $t_4 = 10$. The time-interval from one measurement occasion to the next is constant and equal to 1 hour. In the CT-VAR model, observations can occur at any time between 7 AM and 10 AM for the same four measurement occasions. Specifically, the time points are given by: $t_1 = 7.166667$ (equivalent to 7:10 AM), $t_2 = 8.283333$ (equivalent to 8:17 AM), $t_3 = 9.45$ (equivalent to 9:27 AM), and $t_4 = 10$ (equivalent to 10:00 AM). Notably, the time-interval from one measurement occasion to the next varies allows for irregularly spaced observations. We denote this time-interval as Δt, representing the difference between the current measurement occasion m and the previous measurement occasion $m - 1$. From here we can see that if Δt is constant as in DT-VAR, the model is simply a CT-VAR model where Δt is constant. As such, the DT-VAR model is a special case of the CT-VAR model.

Let us look at the link between the CT-VAR and the discrete-time solution parameters. Whereas, the parameter $\boldsymbol{\Phi}$ describes the moment-to-moment association of one variable on the rate of change in another, the parameter $\boldsymbol{\beta}$ is a matrix of lagged parameters, that is, the effect of the variable on a previous measurement occasion to the current measurement occasion, given a particular time-interval Δt. The discrete-time $\boldsymbol{\beta}$ can be obtained from the continuous-time $\boldsymbol{\Phi}$ using

$$\boldsymbol{\beta} = \exp\left(\Delta t \boldsymbol{\Phi}\right). \tag{6}$$

Furthermore, the discrete-time parameter $\boldsymbol{\alpha}$ is related to the continuous-time parameters $\boldsymbol{\Phi}$ and $\boldsymbol{\mu}$

$$\begin{aligned} \boldsymbol{\alpha} &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\exp\left(\Delta t \boldsymbol{\Phi}\right) - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right) \\ &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\boldsymbol{\beta} - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right). \end{aligned} \tag{7}$$

If we solve for $\boldsymbol{\mu}$ in Equation 7, we arrive at

$$\boldsymbol{\mu} = \left(\boldsymbol{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}. \tag{8}$$

Assuming weak stationarity, the mean vector of $\boldsymbol{\eta}$ as a function of the discrete-time parameters $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ is given by $\boldsymbol{\mu}_{\boldsymbol{\eta}} = \left(\mathbf{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}$. This is equivalent to the CT-VAR parameter $\boldsymbol{\mu}$ as a function of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ given by Equation 8. Lastly, we can convert the continuous-time process noise covariance matrix $\boldsymbol{\Sigma}$ to the discrete-time counter part $\boldsymbol{\Psi}$[3]. Note that $\mathbf{J}$ and $\mathbf{I}$ are identity matrices and $\text{vec}\left(\cdot\right)$ stacks the values from a matrix into a vector by column.

$$\begin{aligned} \text{vec}\left(\boldsymbol{\Psi}\right) &= \mathbf{C}\left[\exp\left(\Delta t \mathbf{C}\right) - \mathbf{J}\right]\text{vec}\left(\boldsymbol{\Sigma}\right) \\ \text{with} \quad \mathbf{C} &= \left(\boldsymbol{\Phi} \otimes \mathbf{I}\right) + \left(\mathbf{I} \otimes \boldsymbol{\Phi}\right) \end{aligned} \tag{9}$$

In summary, we can see the DT-VAR parameters are related to CT-VAR parameters. We can also see from the discrete state space integral solution of CT-VAR

[3] The notation presented here is mathematically equivalent to the notation presented in the works of Arnold (1974), Oud and Jansen (2000), and Voelkle and Oud (2012). We prefer this notation because it facilitates straightforward calculations in statistical packages that support standard linear algebra operations.

that DT-VAR is a special case of CT-VAR where the time-interval Δt is constant across time and individuals. As Δt is part of the calculation of the parameters α, β, and Ψ, their values change as a function of the time-interval.

Quantifying Uncertainty in Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

The work of Deboeck and Preacher (2015) provided a starting point for structuring mediation processes in CT-VAR. However, their work is far from complete: complete inferences require the availability of reasonable standard error estimates and ways of quantifying the uncertainty around the direct, indirect, and total effects in mediation in continuous time. In the mediation literature, it is not enough to have point estimates of the indirect effects. It is important to quantify the uncertainty associated with it. Although the idea of the indirect effect as a product of regression coefficients is straightforward, quantifying the uncertainty of the point estimates has some complications and much of the mediation literature has focused on this task. The sampling distribution of the product of normally distributed random variables is not necessarily normal (Aroian, 1947; Craig, 1936), leading to complexities in computing p-values for the indirect effect. Because of this, confidence intervals have been used to quantify uncertainty for the point estimates. Using the delta method, confidence intervals that are asymptotically normally distributed have been proposed (Aroian, 1947; Goodman, 1960; Sobel, 1982, 1986, 1987). Alternative approaches that take into account the asymmetry have also been proposed including: (1) The distribution of the product method, which analytically examines the sampling distribution of the product of normally distributed coefficients (MacKinnon et al., 2004, 2007); (2) Nonparametric bootstrapping, involving resampling the original data and fitting the model multiple times to create a sampling distribution of the indirect effect (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002); (3) Parametric bootstrapping, involving simulating raw data using the estimated parameters and fitting the model similar to nonparametric bootstrapping (Preacher & Selig, 2012); (4) The Monte

Carlo method, which simulates regression coefficients to generate a sampling distribution of the indirect effect (MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012); (5) Profile likelihood-based approach that estimate confidence intervals using the likelihood ratio test (Cheung, 2009a, 2009b; Pawitan, 2013; Pesigan & Cheung, 2020; Venzon & Moolgavkar, 1988). From these approaches, nonparametric bootstrap has become the preferred method (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002). However, the multiple model fitting especially for more complex models can be a bottleneck making it a less viable approach (see Pesigan & Cheung, 2023, for the case of mediation models with missing data). The Monte Carlo method has been shown to be as good as nonparametric bootstrap in a number of situations (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020). Because the empirical sampling distributions of parameter estimates are generated and not obtained via multiple model fitting, the Monte Carlo method is computationally more efficient than nonparametric bootstrapping.

In this manuscript, drawing from the mediation literature (e.g., MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012; Sobel, 1986), we provide estimation procedures to complete the inferential process in CT-VAR mediation using the delta and Monte Carlo methods. Additionally, we provide the `R` package `cTMed` that calculates point estimates, standard errors, and confidence intervals for the direct, indirect, and total effects. The package also provides plotting tools to visualize how the effects change given a range of time-intervals.

References

Aalen, O. O., Røysland, K., Gran, J. M., Kouyos, R., & Lange, T. (2016). Can we believe the DAGs? a comment on the relationship between causal DAGs and mechanisms. *Statistical Methods in Medical Research*, *25*(5), 2294–2314. https://doi.org/10.1177/0962280213520436

Aalen, O. O., Røysland, K., Gran, J. M., & Ledergerber, B. (2012). Causality, mediation and time: A dynamic viewpoint. *Journal of the Royal Statistical Society. Series A (Statistics in Society)*, *175*(4), 831–861. http://www.jstor.org/stable/23355305

Arnold, L. (1974). *Stochastic differential equations: Theory and applications: Theory and applications*. Wiley.

Aroian, L. A. (1947). The probability function of the product of two normally distributed variables. *The Annals of Mathematical Statistics*, *18*(2), 265–271. https://doi.org/10.1214/aoms/1177730442

Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, *51*(6), 1173–1182. https://doi.org/10.1037/0022-3514.51.6.1173

Boker, S. M. (2002). Consequences of continuity: The hunt for intrinsic properties within parameters of dynamics in psychological processes. *Multivariate Behavioral Research*, *37*(3), 405–422. https://doi.org/10.1207/s15327906mbr3703_5

Bollen, K. A., & Stine, R. (1990). Direct and indirect effects: Classical and bootstrap estimates of variability. *Sociological Methodology*, *20*, 115. https://doi.org/10.2307/271084

Cheung, M. W.-L. (2009a). Comparison of methods for constructing confidence intervals of standardized indirect effects. *Behavior Research Methods*, *41*(2), 425–438. https://doi.org/10.3758/brm.41.2.425

Cheung, M. W.-L. (2009b). Constructing approximate confidence intervals for parameters with structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *16*(2), 267–294. https://doi.org/10.1080/10705510902751291

Chow, S.-M., Losardo, D., Park, J., & Molenaar, P. C. M. (2023). Continuous-time dynamic models: Connections to structural equation models and other discrete-time models. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (2nd ed.). The Guilford Press.

Cole, D. A., & Maxwell, S. E. (2003). Testing mediational models with longitudinal data: Questions and tips in the use of structural equation modeling. *Journal of Abnormal Psychology*, *112*(4), 558–577. https://doi.org/10.1037/0021-843x.112.4.558

Craig, C. C. (1936). On the frequency function of xy. *The Annals of Mathematical Statistics*, *7*(1), 1–15. https://doi.org/10.1214/aoms/1177732541

de Haan-Rietdijk, S., Voelkle, M. C., Keijsers, L., & Hamaker, E. L. (2017). Discrete- vs. continuous-time modeling of unequally spaced experience sampling method data. *Frontiers in Psychology*, *8*. https://doi.org/10.3389/fpsyg.2017.01849

Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis. *Structural Equation Modeling: A Multidisciplinary Journal*, *23*(1), 61–75. https://doi.org/10.1080/10705511.2014.973960

Fairchild, A. J., & MacKinnon, D. P. (2014). Using mediation and moderation analyses to enhance prevention research. In *Defining prevention science* (pp. 537–555). Springer US. https://doi.org/10.1007/978-1-4899-7424-2_23

Georgeson, A. R., Alvarez-Bartolo, D., & MacKinnon, D. P. (2023). A sensitivity analysis for temporal bias in cross-sectional mediation. *Psychological Methods*. https://doi.org/10.1037/met0000628

Gollob, H. F., & Reichardt, C. S. (1987). Taking account of time lags in causal models. *Child Development*, *58*(1), 80. https://doi.org/10.2307/1130293

Gollob, H. F., & Reichardt, C. S. (1991). Interpreting and estimating indirect effects assuming time lags really matter. In L. M. Collins & J. L. Horn (Eds.), *Best methods for the analysis of change: Recent advances, unanswered questions, future directions* (pp. 243–259). American Psychological Association. https://doi.org/10.1037/10099-015

Goodman, L. A. (1960). On the exact variance of products. *Journal of the American Statistical Association*, *55*(292), 708–713. https://doi.org/10.1080/01621459.1960.10483369

Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, *37*(3), 424. https://doi.org/10.2307/1912791

James, L. R., & Brett, J. M. (1984). Mediators, moderators, and tests for mediation. *Journal of Applied Psychology*, *69*(2), 307–321. https://doi.org/10.1037/0021-9010.69.2.307

Judd, C. M., & Kenny, D. A. (1981). Process analysis. *Evaluation Review*, *5*(5), 602–619. https://doi.org/10.1177/0193841x8100500502

MacKinnon, D. P. (1994). Analysis of mediating variables in prevention and intervention research. *NIDA research monograph*, *139*, 127–153.

MacKinnon, D. P. (2008). *Introduction to statistical mediation analysis*. Erlbaum Psych Press. https://doi.org/10.4324/9780203809556

Mackinnon, D. P., & Dwyer, J. H. (1993). Estimating mediated effects in prevention studies. *Evaluation Review*, *17*(2), 144–158. https://doi.org/10.1177/0193841x9301700202

MacKinnon, D. P., Fritz, M. S., Williams, J., & Lockwood, C. M. (2007). Distribution of the product confidence limits for the indirect effect: Program PRODCLIN. *Behavior Research Methods*, *39*(3), 384–389. https://doi.org/10.3758/bf03193007

MacKinnon, D. P., Lockwood, C. M., & Williams, J. (2004). Confidence limits for the indirect effect: Distribution of the product and resampling methods. *Multivariate Behavioral Research*, *39*(1), 99–128. https://doi.org/10.1207/s15327906mbr3901_4

Maxwell, S. E., & Cole, D. A. (2007). Bias in cross-sectional analyses of longitudinal mediation. *Psychological Methods*, *12*(1), 23–44. https://doi.org/10.1037/1082-989x.12.1.23

Maxwell, S. E., Cole, D. A., & Mitchell, M. A. (2011). Bias in cross-sectional analyses of longitudinal mediation: Partial and complete mediation under an autoregressive model. *Multivariate Behavioral Research*, *46*(5), 816–841. https://doi.org/10.1080/00273171.2011.606716

O'Laughlin, K. D., Martin, M. J., & Ferrer, E. (2018). Cross-sectional analysis of longitudinal mediation processes. *Multivariate Behavioral Research*, *53*(3), 375–402. https://doi.org/10.1080/00273171.2018.1454822

Oravecz, Z., Tuerlinckx, F., & Vandekerckhove, J. (2011). A hierarchical latent stochastic differential equation model for affective dynamics. *Psychological Methods*, *16*(4), 468–490. https://doi.org/10.1037/a0024375

O'Rourke, H. P., & MacKinnon, D. P. (2018). Reasons for testing mediation in the absence of an intervention effect: A research imperative in prevention and intervention research. *Journal of Studies on Alcohol and Drugs*, *79*(2), 171–181. https://doi.org/10.15288/jsad.2018.79.171

O'Rourke, H. P., & MacKinnon, D. P. (2019). The importance of mediation analysis in substance-use prevention. In *Advances in prevention science* (pp. 233–246). Springer International Publishing. https://doi.org/10.1007/978-3-030-00627-3_15

Oud, J. H. L., & Delsing, M. J. M. H. (2010). Continuous time modeling of panel data by means of SEM. In K. van Montfort, J. H. L. Oud, & A. Satorra (Eds.), *Longitudinal research with latent variables* (pp. 201–244). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-11760-2_7

Oud, J. H. L., & Jansen, R. A. R. G. (2000). Continuous time state space modeling of panel data by means of SEM. *Psychometrika*, *65*(2), 199–215. https://doi.org/10.1007/bf02294374

Pawitan, Y. (2013). *In all likelihood: Statistical modelling and inference using likelihood*. Oxford University Press.

Pesigan, I. J. A., & Cheung, S. F. (2020). SEM-based methods to form confidence intervals for indirect effect: Still applicable given nonnormality, under certain conditions. *Frontiers in Psychology*, *11*. https://doi.org/10.3389/fpsyg.2020.571928

Pesigan, I. J. A., & Cheung, S. F. (2023). Monte Carlo confidence intervals for the indirect effect with missing data. *Behavior Research Methods*. https://doi.org/10.3758/s13428-023-02114-4

Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, *40*(3), 879–891. https://doi.org/10.3758/brm.40.3.879

Preacher, K. J., & Selig, J. P. (2012). Advantages of Monte Carlo confidence intervals for indirect effects. *Communication Methods and Measures*, *6*(2), 77–98. https://doi.org/10.1080/19312458.2012.679848

Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, *87*(1), 214–252. https://doi.org/10.1007/s11336-021-09767-0

Shrout, P. E., & Bolger, N. (2002). Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods*, *7*(4), 422–445. https://doi.org/10.1037/1082-989x.7.4.422

Sobel, M. E. (1982). Asymptotic confidence intervals for indirect effects in structural equation models. *Sociological Methodology*, *13*, 290. https://doi.org/10.2307/270723

Sobel, M. E. (1986). Some new results on indirect effects and their standard errors in covariance structure models. *Sociological Methodology*, *16*, 159. https://doi.org/10.2307/270922

Sobel, M. E. (1987). Direct and indirect effects in linear structural equation models. *Sociological Methods & Research*, *16*(1), 155–176. https://doi.org/10.1177/0049124187016001006

Tofighi, D., & Kelley, K. (2019). Indirect effects in sequential mediation models: Evaluating methods for hypothesis testing and confidence interval formation. *Multivariate Behavioral Research*, *55*(2), 188–210. https://doi.org/10.1080/00273171.2019.1618545

Tofighi, D., & Kelley, K. (2020). Improved inference in mediation analysis: Introducing the model-based constrained optimization procedure. *Psychological Methods*, *25*, 496–515. https://doi.org/10.1037/met0000259

Uhlenbeck, G. E., & Ornstein, L. S. (1930). On the theory of the brownian motion. *Physical Review*, *36*(5), 823–841. https://doi.org/10.1103/physrev.36.823

van Montfort, K., Oud, J. H. L., & Voelkle, M. C. (Eds.). (2018). *Continuous time modeling in the behavioral and related sciences*. Springer International Publishing. https://doi.org/10.1007/978-3-319-77219-6

Venzon, D. J., & Moolgavkar, S. H. (1988). A method for computing profile-likelihood-based confidence intervals. *Applied Statistics*, *37*(1), 87. https://doi.org/10.2307/2347496

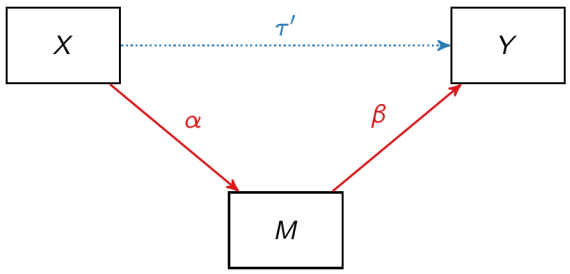
Voelkle, M. C., & Oud, J. H. L. (2012). Continuous time modelling with individually varying time intervals for oscillating and non-oscillating processes. *British Journal of Mathematical and Statistical Psychology*, *66*(1), 103–126. https://doi.org/10.1111/j.2044-8317.2012.02043.x

Voelkle, M. C., Oud, J. H. L., Davidov, E., & Schmidt, P. (2012). An SEM approach to continuous time modeling of panel data: Relating authoritarianism and anomia. *Psychological Methods*, *17*(2), 176–192. https://doi.org/10.1037/a0027543

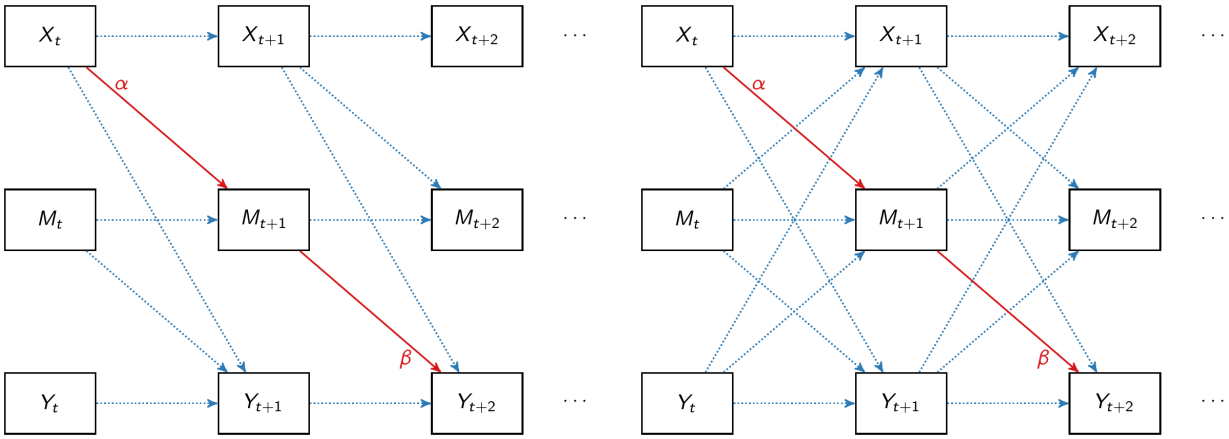
Wang, L., & Zhang, Q. (2020). Investigating the impact of the time interval selection on autoregressive mediation modeling: Result interpretations, effect reporting, and temporal designs. *Psychological Methods*, *25*(3), 271–291. https://doi.org/10.1037/met0000235

Figure 1

Cross-Sectional and Longitudinal Mediation Models



(a) *Cross-sectional mediation model.*



(b) *Unidirectional autoregressive cross-lagged mediation model.*

(c) *Bidirectional autoregressive cross-lagged mediation model.*

Note: The indirect effect of M on the path from X to Y is the product of α and β. t, $t+1$, and $t+2$ represent equally spaced time points. The dots indicate that the model can accommodate more time points. In the bidirectional autoregressive cross-lagged mediation model, the path from the dependent variable to the independent and the mediator variables as well as the path from the mediator variable to the independent are included in the model. These paths are omitted in the unidirectional model.